# WAVE MOTION

Waves transmit energy from one point to another in a substance (called a medium) without the particles of the substance moving with it.

This leads to the following definition.

| |
|---|
| ***Wave motion***: a disturbance or vibration that transmits energy without the movement of matter from one point to another. |

Waves that require a medium are called ***mechanical waves***.

e.g. Sound, water waves, waves in strings and springs.

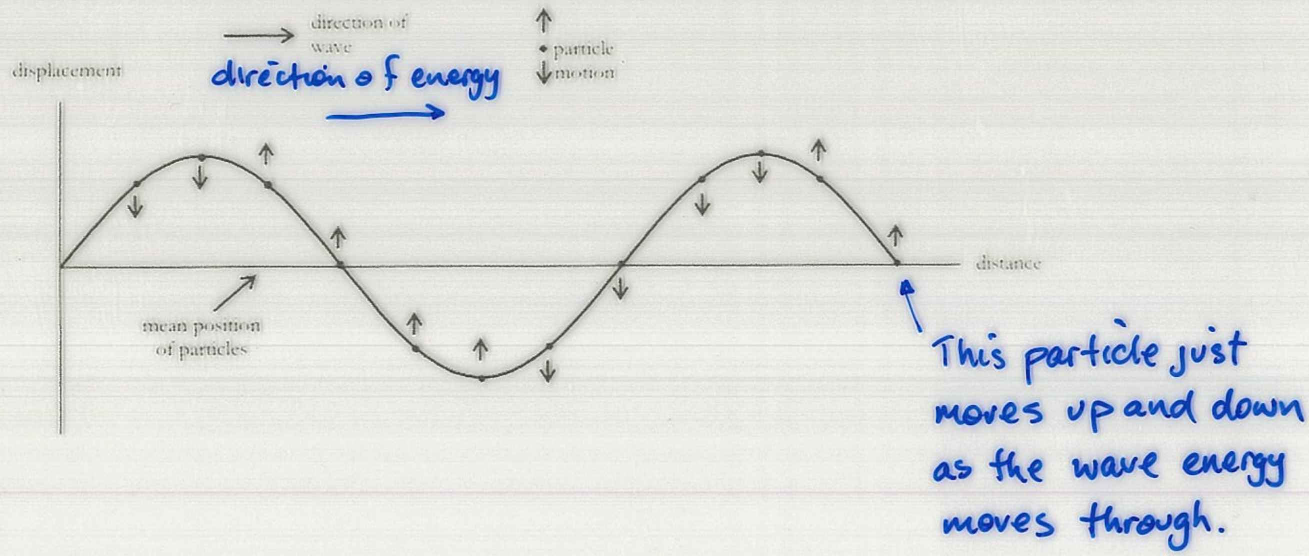
Waves that don't need a medium are called ***electromagnetic waves***.

e.g. Light.

There are two types of mechanical waves.

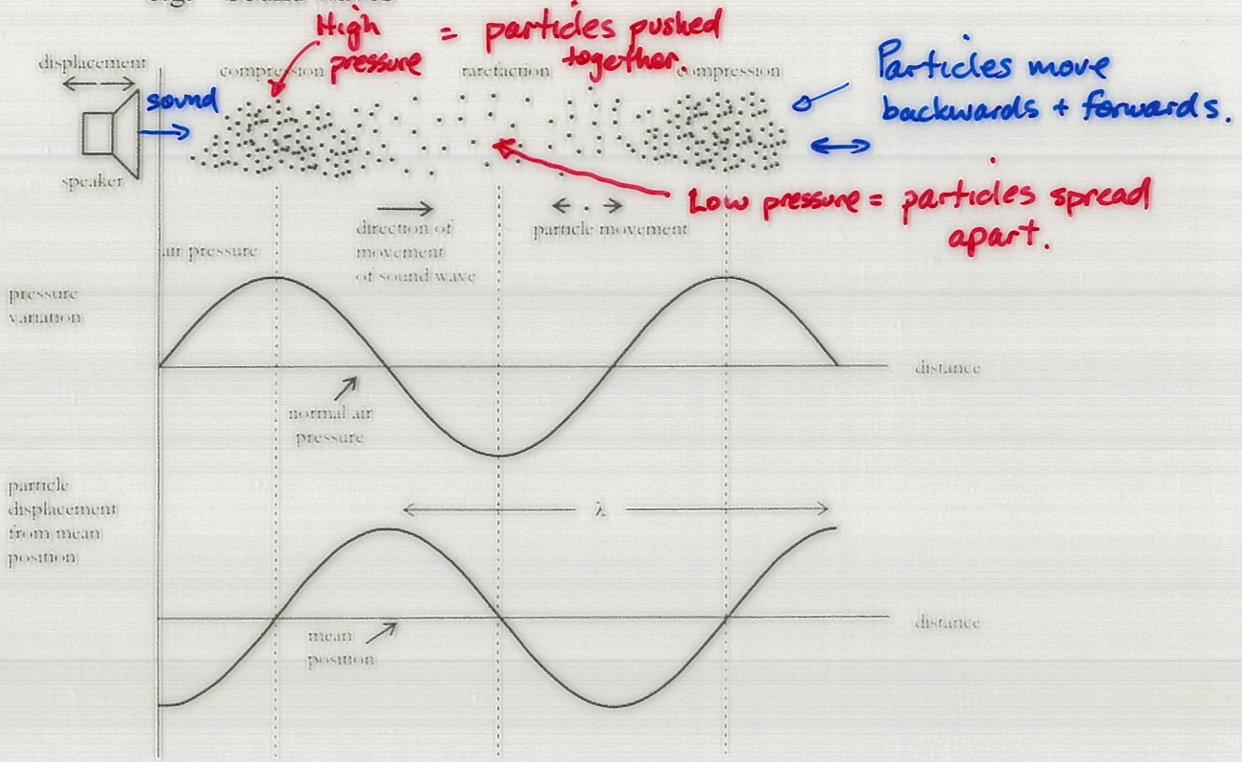
**Transverse wave** - particles move ***perpendicular (up and down)*** at right angles to the direction of the movement of the wave.

e.g. Water waves.



**Longitudinal wave** - particles move ***parallel (back and forward)*** in the same direction as the direction of the movement of the wave.

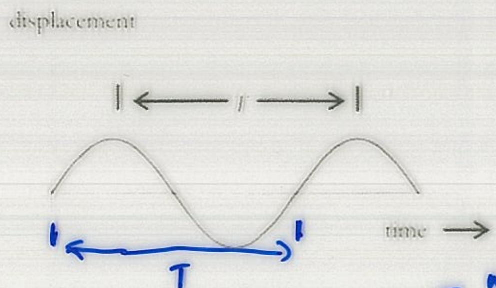
e.g. Sound waves




**Wave graphs**

Wave motion can be represented by a displacement/ time graph or a displacement/distance graph.

- **Displacement / Time**



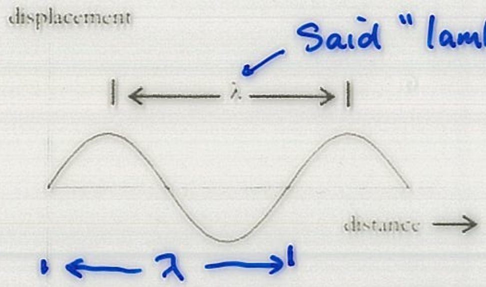

moves up + down, or backwards + forwards.

This graph represents the ***motion of a single particle*** as a wave passes through it.

The crests and troughs represent the particle's maximum displacement from its mean position.

***T is the period of vibration*** - time taken for one complete oscillation (up-down and back to start position).

- **Displacement / Distance** Like taking a photograph.



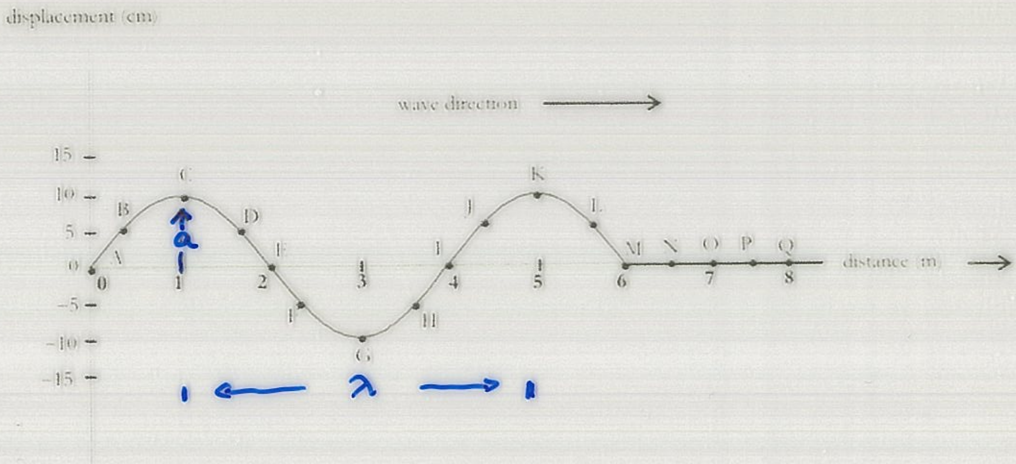

Shows all the particles of the substance (medium)

This graph represents the ***movement of a number of particles*** from their mean position at some instant in time.

Think of it as a "snapshot" or photograph that shows how the displacement of the particles of the medium varies over the distance at that instant.

***λ represents the wavelength of the wave***.

## Describing Wave Motion - Important Terms




**Wavelength** (λ) The distance between any two crests or troughs.

e.g. The distance between points C & K, A & I, etc.

**Amplitude** (a) The maximum displacement of a particle from its mean (normal) position.

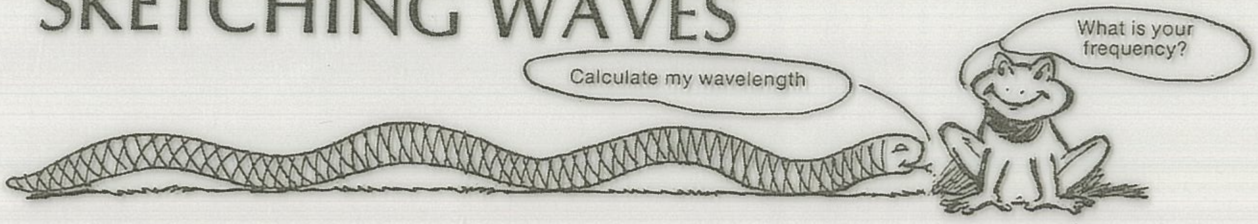
i.e. Height of a crest or depth of a trough.
The amplitude of the wave shown is 10 cm.

**Frequency** (f) The number of complete vibrations (or waves) per second.

The unit of frequency is the hertz (Hz).

**Period** ( T) The time taken to complete one vibration (or to produce one complete wave).

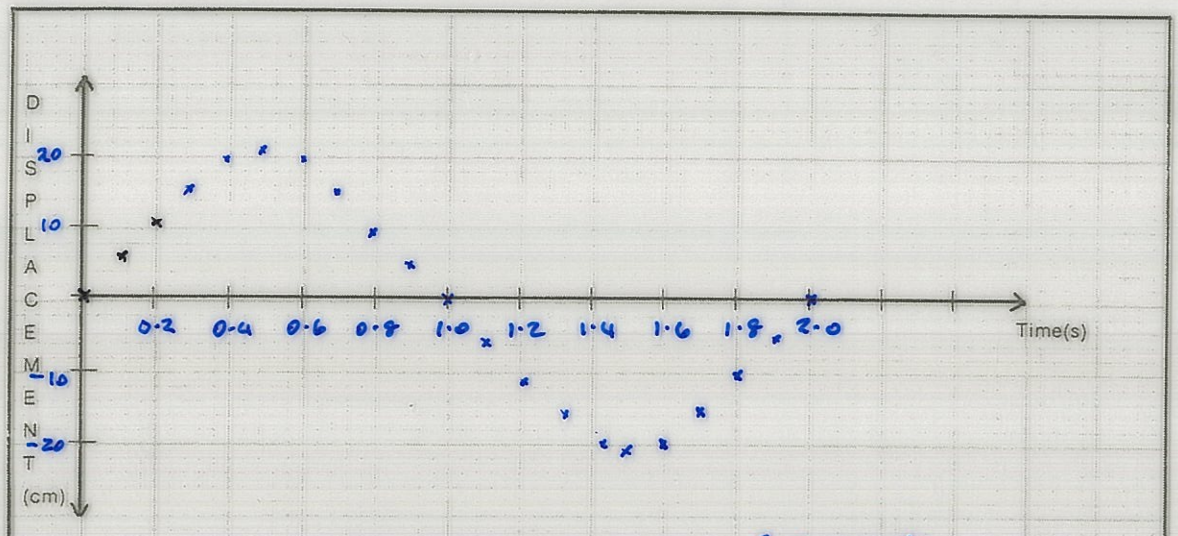
# SKETCHING WAVES




The following data have been obtained by photo-analysis of the motion of a single particle in a medium as energy in a wave form passes through the medium.

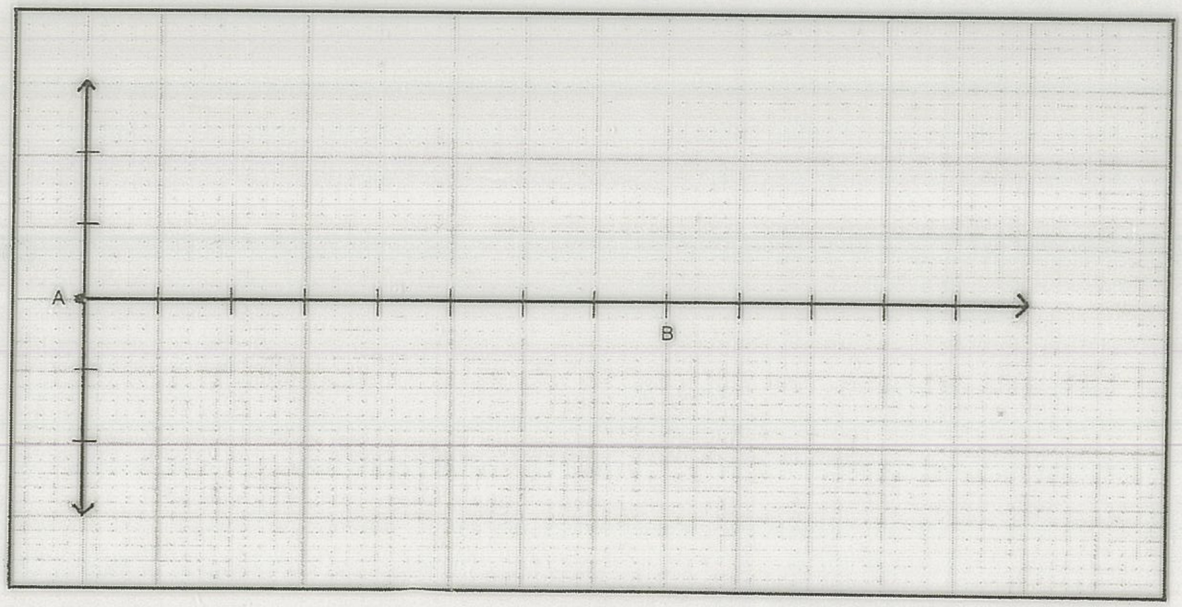
| Time (s) | 0.0 | 0.1 | 0.2 | 0.3 | 0.4 | 0.5 | 0.6 | 0.7 | 0.8 | 0.9 | 1.0 | 1.1 | 1.2 | 1.3 | 1.4 | 1.5 | 1.6 | 1.7 | 1.8 | 1.9 | 2.0 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Displacement from rest position (cm) | 0 | 5 | 11 | 16 | 19 | 20 | 19 | 16 | 11 | 5 | 0 | -5 | -11 | -16 | -19 | -20 | -19 | -16 | -11 | -5 | 0 |

Plot the data on the graph paper.




What is the time for one complete cycle? 2·0 seconds

Using the piece of wire you have been given, construct the outline of a wave with a wavelength of 4 cm and an amplitude of 3 cm. Trace the outline onto the graph paper.




If the wave takes 1 second to travel from A to B, calculate the frequency.

# SOUND AND WAVES

## *Background*

Sound is a form of wave motion that we use every day. What would our world be like if it were silent? Wave motion has various characteristics that need to be understood and learned.

## *Task*

Your task is to use the Internet to find useful information that can provide ***simple answers*** to the following questions. Each set of questions has a website that should be visited first.

You may find others by using Metacrawler (http://www.metacrawler.com/) and doing a global search on waves, sound, or something similar. Applets are interactive and may be very useful.

## *Questions*

**Go to the following website.**

http://www.ngsir.netfirms.com/englishVersion.htm

- Click on Light and Wave.
- Select Transverse Wave (run).
- Select Start.
- Choose the Select a Point button.

1. How does the selected point move?

2. What happens when you change the following?

   (i) Amplitude

   (ii) Wavelength

   (iii) Speed

- Go back one page.
- Select Longitudinal Travelling Wave (run).
- Choose the Select a Point button.

3. How does the selected point move?

- Select the Graph button - the movement now shows as a graph.

4. Draw what the shape of the graph looks like.

5. What happens when you change the following?

   (i) Amplitude

   (ii) Wavelength

   (iii) Speed

**Go to the following website.**

http://www.surendranath.org/Applets.html

- Select Applet Menu
- Choose Waves.
- Select Transverse Waves.

6. What happens when you change the amplitude and frequency?

- Repeat for Longitudinal Waves.

**Go to the following website.**

http://www.fi.edu/fellows/fellow2/apr99/soundindex.html

- Click on How Sound is Made.

7. What are sound waves?

8. In which direction do the particles of the medium move as a sound moves through it?

9. Draw a simple diagram to show how a tuning fork produces a sound.

10. What is a:

    (i) compression?

    (ii) rarefaction?

11. Can a sound wave travel through a vacuum? Briefly explain your answer.

**Go back 1 page.**

- Click on Music to Our Ears.
- Click on The Ear.

12. What do the following parts of the ear do?

(i) Outer ear. Collects sound.

(ii) Ear drum. Vibrates with the sound.

(iii) Hammer, anvil and stirrup (in the Middle Ear). Transfer vibrations to the cochlea ("snail shell

(iv) Cochlea. Changes vibrations into nerve signals.

(v) Auditory nerve. Carries the nerve signals to the brain.

13. Do sound waves enter our brain? What does?

No - only nerve signals.

**Go back 2 pages.**

- Click on Good Vibrations.

14. On what device can we see a picture of a sound wave? Cathode ray oscilloscope (CRO).

15. Define the following.

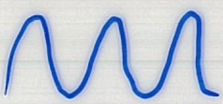
(i) Wavelength Distance from crest to crest

(ii) Frequency Number of waves per second

(iii) Amplitude Height of the wave - a measure of the energy in a sound wave.

16. Draw a graph to show the difference between the following.

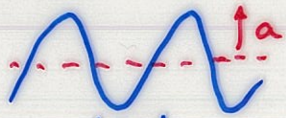
(i) A high pitch and a low pitch sound. Pitch = frequency. (How it sounds!)
e.g. High pitch = high frequency.



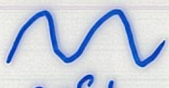
High pitch

Low pitch

(ii) A loud and soft sound.
↑ amplitude



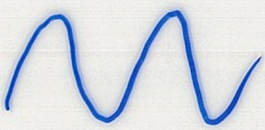

loud



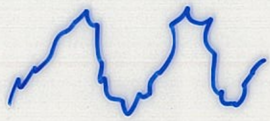
soft

Same frequency (pitch)

(iii) A pleasant sound and noise.



Pure note / sound



Noise

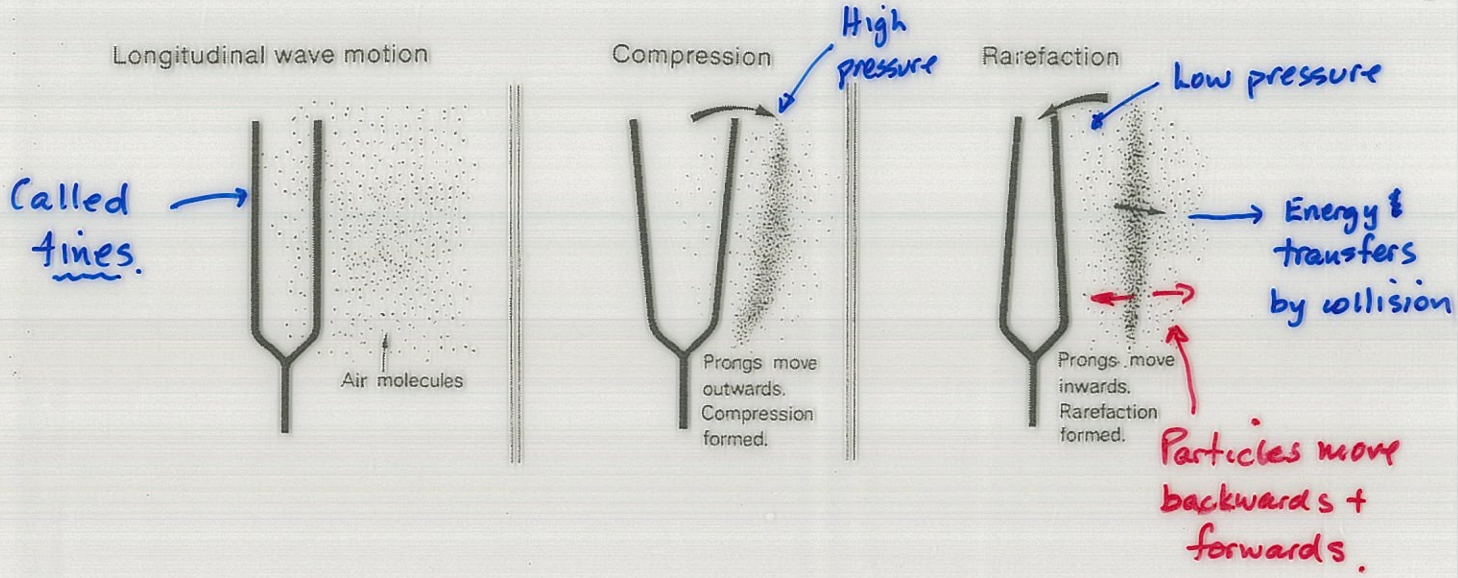
# PRODUCTION OF SOUND

Sound waves are produced by a **vibrating source**.

a substance (solid, liquid, gas)

They need a **medium** to travel in, such as air, water or a solid. Sound cannot be heard in space.

e.g. A tuning fork.




The sound waves produced are **longitudinal waves** made up of a series of **compressions** (high pressure areas) and **rarefactions** (low pressure areas).

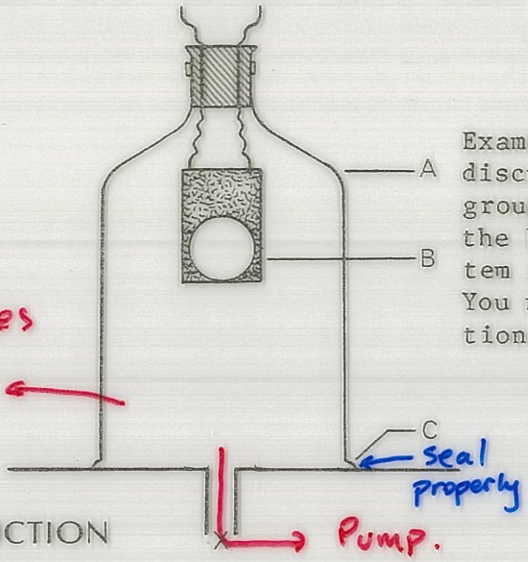
**Energy** is transferred through the medium without the particles moving from one place to another.

The **frequency of vibration** determines the sound that is heard - a high frequency sounds high, a low frequency sounds low.

e.g. Jet engines have a high frequency whine while ceiling fans have a low hum.

# HOW SOUND TRAVELS

Examine the diagram and, after some discussion with other members of your group, predict what will happen when the bell is switched on and the system is connected to a vacuum pump. You may like to support your prediction with some reasons.




Air pressure goes lower - still has some air inside.

A Bell jar
B Electric buzzer
C Petroleum jelly seal
X To vacuum pump

## GROUP PREDICTION

## REASON

Your teacher has set up some similar equipment which will allow the class to observe what actually happens.

## OBSERVATIONS

- Could still hear the bell.
- Same loudness - no
  Softer - yes.

## CONCLUSIONS

1. Did the results from the demonstration support your group's prediction?

2. Can you suggest an explanation for the experimental results?
   - Goes softer since there is less air.
   - Not as much air to transfer energy.

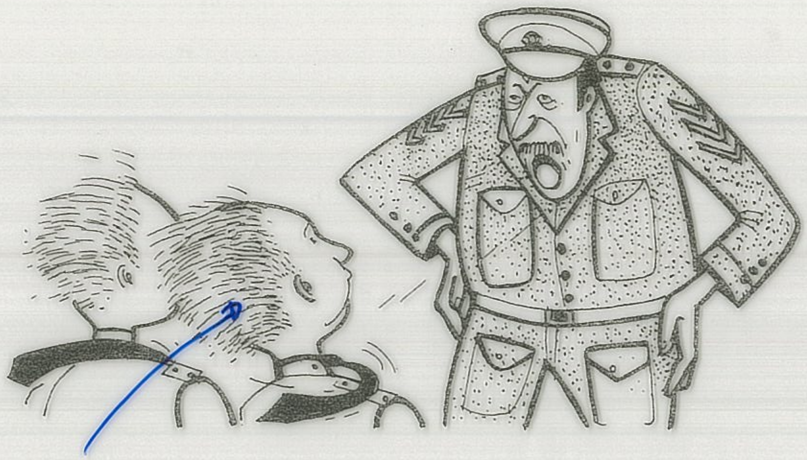
3. Make some suggestions as to how this experiment could be improved.
   - Need more air out - get a better pump.
   - Seal the jar properly.

4. From this activity, can you suggest what is needed before sound can be heard?
   - Air
   - Solid or liquid

   } • Need a substance (medium)

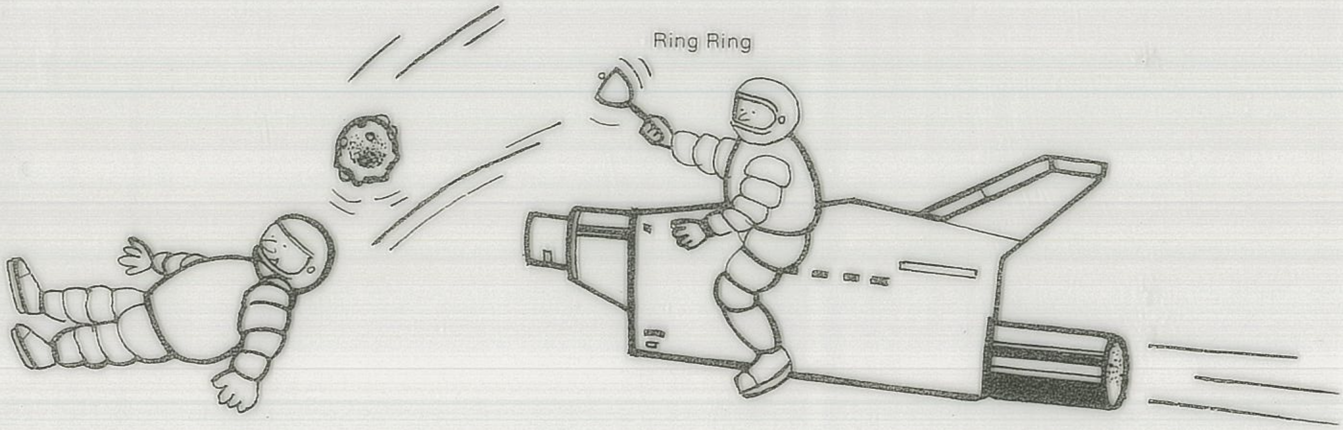
# MORE NOISE

DISCUSS these diagrams and try to work out what is wrong.



SPOT THE ERROR Hair won't move

This cannot happen because air particles don't move from the guy's mouth to the boys' hair



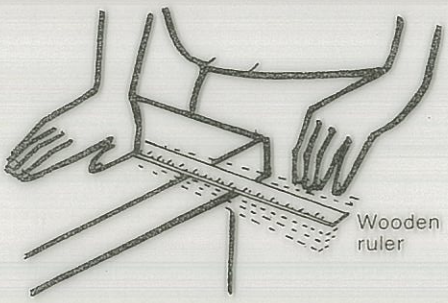

SPOT THE ERROR Sound of ringing.

This cannot happen because no air in space.

# PRODUCE THE FOLLOWING SOUNI


PROCEDURE
Cut the bottom out of a plastic cup, then stretch some adhesive food wrapping over the cut end of the cup. Speak into the open end of the cup and record your observations.

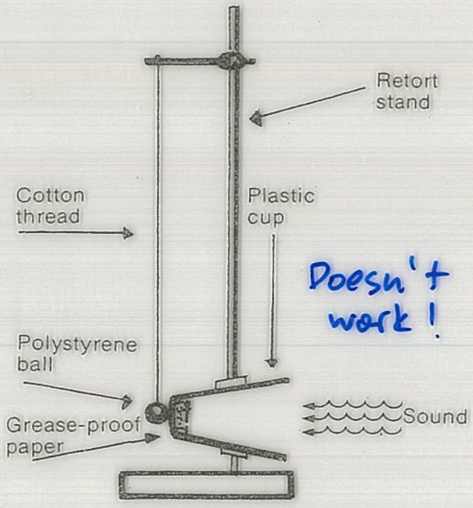
OBSERVATIONS: Should vibrate.




Place a ruler on the top of the bench and have 15 to 20 cm overhanging the edge. Holding the ruler firmly on the desk with one hand, pluck the free end and record your observations.

OBSERVATIONS: Long ruler → low sound

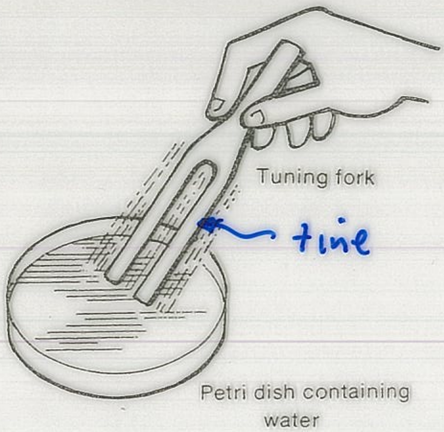
Short ruler - high sound




Set up the equipment as shown in the diagram, with the small polystyrene ball resting against the centre of the paper. (*Note: the thread is not quite vertical.*) Before you speak into the cup, make a prediction.
I predict Should vibrate the ball.

Test your prediction.

Explain the result.




Sound a tuning fork by striking the prongs on a *soft* surface. While the fork is vibrating, gently touch the prongs on the surface of the water. Record your observations.

OBSERVATIONS: Water vibrates out of the container.

Can you explain your observations?

Tines vibrate

BRIEFLY SUMMARIZE YOUR OBSERVATIONS

To make sound, we need something vibrating.

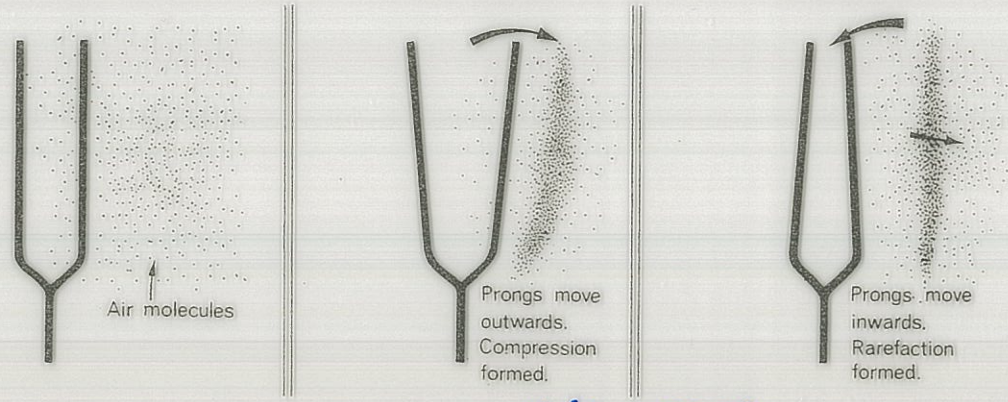
# HOW SOUND TRAVELS THROUGH A GAS

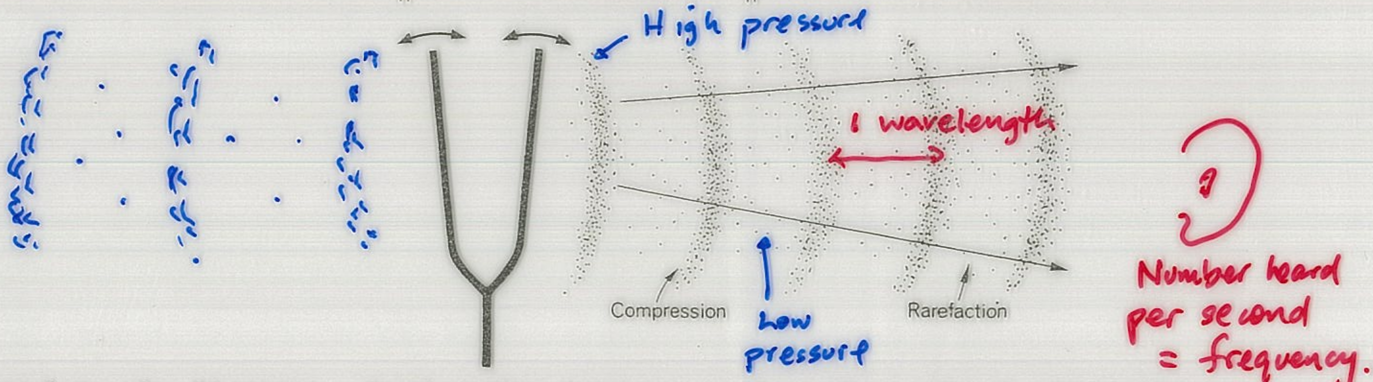
The incomplete diagrams below show how sound travels through a gas. Copy the diagrams into your notes. When you have done this, complete the diagrams you have copied. You will need to use a textbook to find out more about how sound travels through a gas.

To help, some key words have been listed.

## KEY WORDS

Longitudinal wave motion    Compression    Rarefaction




Complete the diagrams, showing what happens on the other side of the tuning fork. You must be sure to label the compressions and rarefactions formed. Beneath the diagrams write a brief paragraph describing, in your own words, how sound travels through a gas.

## BRAIN TEASERS

TRY THESE! Sound energy travels quickly - gas particle travel slowly.

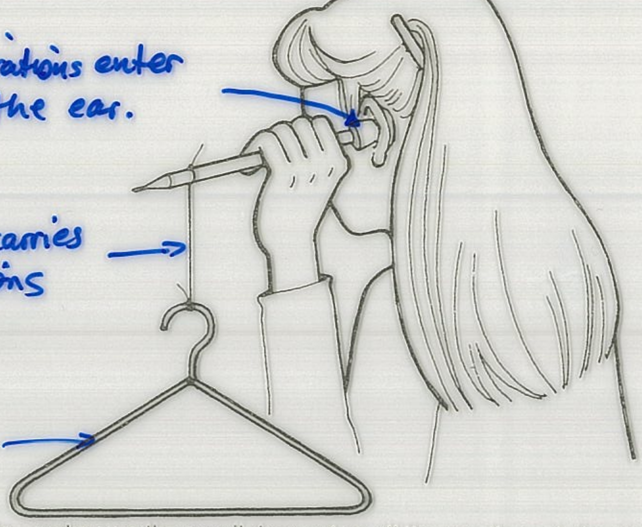
1. A gas tap is opened on the front desk. The sound of the escaping gas is heard almost instantly at all parts of the room. After some time, the smell of the gas is detected. What does this show?

2. Explain why a sound becomes softer as you move away from its source.
Energy spreads out so sound is softer.

3. Why is it difficult to hear or to make yourself heard when shouting into the wind? Wind carries the sound energy away.

4. Sound can be heard over longer distances after it has been raining. Why do you think this occurs?
Moisture in air conducts the sound better.

# SOUNDS IN SOLIDS




Note—make sure the pencil does not go all the way through the stopper.

## PROCEDURE

1. Carefully place the stoppered end of the pencil to your ear.

2. Hit the coat hanger with another pencil.

## RESULTS

1. What do you hear with the ear to which the stoppered end of the pencil is placed? Rings clearly.

2. What was the source of the sound? Knocking the metal.

3. Why didn't you hear the coat hanger vibrations through the air?
Not a lot of energy - it spread out.

4. Describe how the vibrations reached the ear from the source.

## EXPLAIN THESE

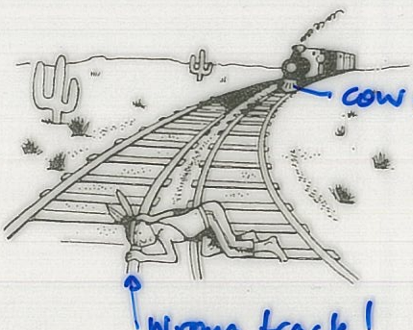
Iron rod carries the vibrations from the engine to the ear.

1. A 'knock' in a car engine can be located by putting one end of an iron rod here and there on the engine block and holding the other end near your ear.

2. If an object were struck under water, would a nearby diver hear a sound? Yes - easily. Water conducts sound much better than air.

3. Explain how people trapped in a mine cave-in could be located.
Make noise to be heard.

4. Write a brief description of what is happening in the diagram.



# SPEED OF SOUND

Sound travels fastest in a solid, slowest in a gas.

The particles are much closer together in a solid so they collide with each other very quickly so the sound energy is transferred much faster than in a liquid or gas.

| MEDIUM | SPEED OF SOUND |
|---|---|
| air | 340 $ms^{-1}$ |
| water | ~ 4 times faster than air |
| solid | ~ 15 times faster than air |

To calculate the speed of sound:

$$\textbf{Speed of sound} = \frac{\textbf{distance travelled}}{\textbf{time taken}}$$

**EXAMPLE**

The sound of thunder is heard 4.8 seconds after the flash of lightning. If the velocity of sound is 340 $ms^{-1}$, how far was the lightning away from the observer?

$$\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{d}{t}$$

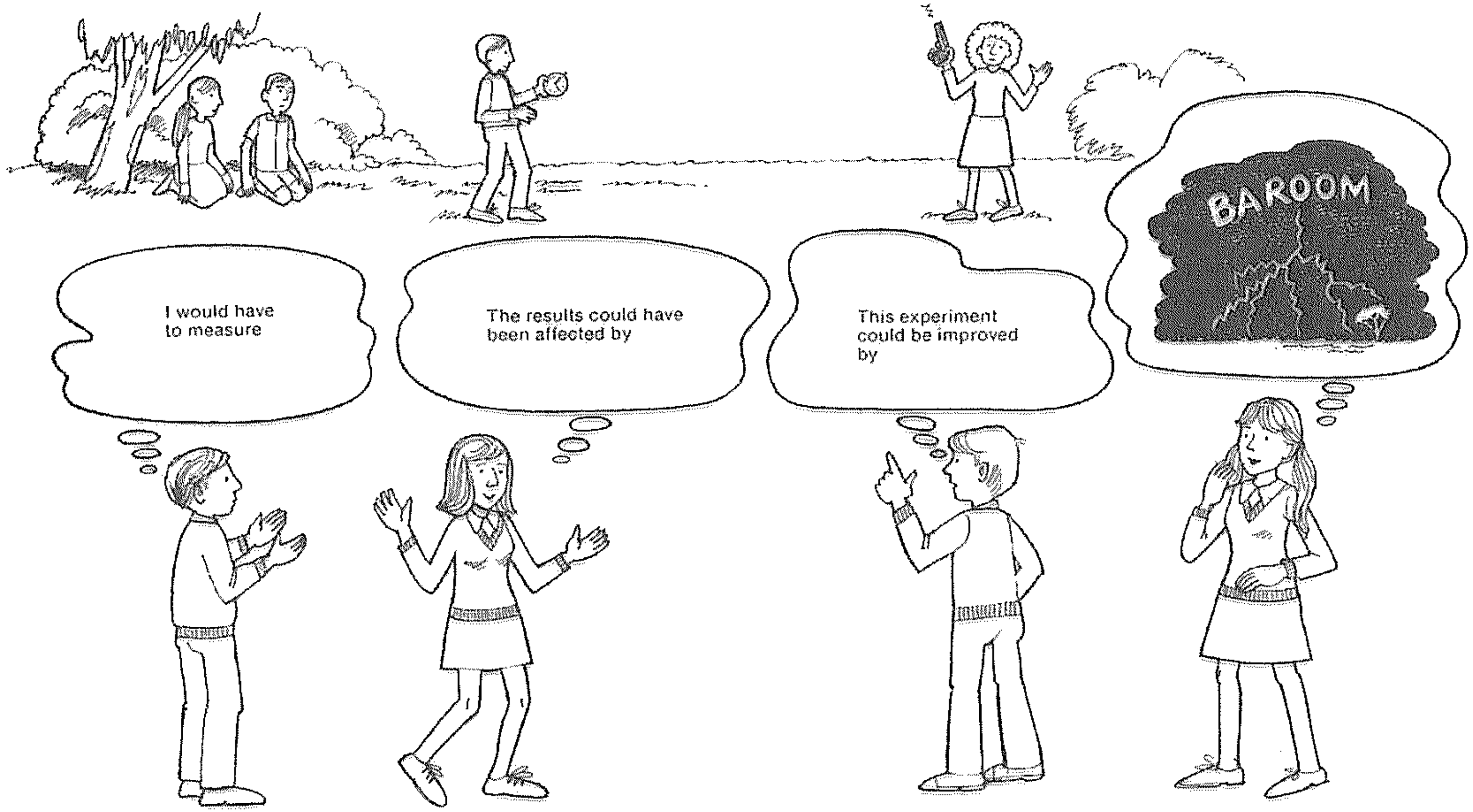
$$\Rightarrow \quad 340 = \frac{d}{4.8}$$

$$\Rightarrow \quad d = (340)(4.8)$$

$$= 1632 \text{ m}$$

$\therefore$ The lightning was 1632 m away.

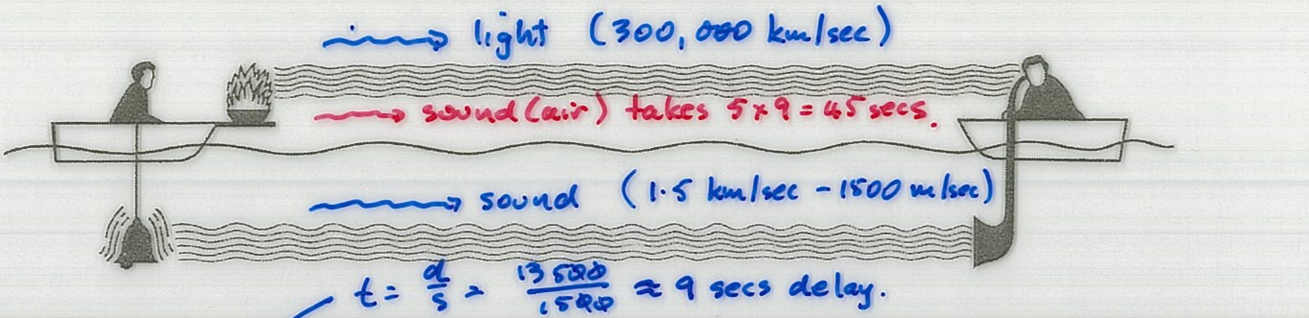
# HOW CAN YOU MEASURE THE SPEED OF SOUND?




## YOU COULD CARRY OUT A SIMILAR EXPERIMENT

1. What differences between the properties of sound and light are used in this experiment?
2. Why did this experiment make the person think of thunder and lightning?

# The Colladon Experiment




In 1826, a historic experiment was carried out on Lake Geneva. Two boats were moored 13 500 metres apart. One carried a 70 kg bell (below the water) and a small tray of gunpowder.

The gunpowder was ignited just as a hammer struck the bell. An observer in the second boat was able to see the flash from the gun-powder and, by using an ear trumpet, to hear the sound of the bell. He started his stop-watch when he saw the flash and stopped it when he heard the sound in the ear trumpet.

1. What was being measured in this experiment? Do you think the results would be accurate? Speed of sound in water. Accurate - yes - large time delay.

2. How did the distance between the boats affect the time it took for the listener to hear the explosion? Did this affect the result of the experiment? Explain. Further apart mean bigger time delay to hear the explosion.

3. The observer heard two explosions. Explain why this happened.
   Hear the one in water first, then the one in air second.

4. What effect would atmospheric conditions, such as wind speed and direction or air temperature, have had on the accuracy of the result? Explain. No effect - listening in the water.

5. Explain how the following affected the experimental results:

   (a) The observer's reaction time. Yes but it's very small.

   (b) The depth of the bell in the water. No.

   (c) The purity of the water. Yes - salty water has a higher speed.

   (d) The temperature of the water. Yes - it can speed up or slow down.

6. The scientists repeated the experiment many times before they published their results. Explain why they collected so many results.

   More times = more accurate.